The Cost of Reliable Near-Optimal Set Maintenance with Arriving Arms

Deqi Zheng* Xiaoyang Xu*
Qiuzhen College, Tsinghua University

Yuhong Yang†
Yau Mathematical Sciences Center, Tsinghua University

October 15, 2026

Abstract

We study the cost of maintaining reliable reports of all relatively near-optimal arms while collecting rewards when new arms arrive and remain available. Reward laws are stationary, but each arrival can change the membership boundary. We measure reporting progress by cumulative unresolved arm time outside a prescribed boundary band, under simultaneous correctness of all issued labels. Uncertainty about one new arm can obstruct positive certification of many earlier candidates. A lower bound in one common reference environment quantifies this effect. We obtain matching worst-schedule orders for unresolved area and an arrival-growth boundary for jointly sublinear area and expected regret. On a specified repeated-update family, rapid maintenance requires inverse-square regret, whereas a constant-factor larger area budget admits a logarithmic regret upper bound. The rapid cost also holds with nonvanishing probability in a fixed environment. A fixed-precision checkpoint construction provides the upper bounds without knowing future arrivals. Finite-budget comparisons show that its conservative certification cost can be substantial; they do not establish uniform numerical superiority or a general optimal regret–area frontier.

Keywords: sequential experimentation; arriving arms; near-optimal set maintenance; dynamic regret; simultaneous inference.

1 Introduction

Sequential experiments often acquire new candidates while earlier ones remain available. An allocation rule must learn which candidates yield high rewards. An experimenter may also need to know which of all available candidates are within a specified tolerance of the current best. These objectives need not favor the same observations. A candidate can be sufficiently inferior to receive

*The first two authors contributed equally.

†Corresponding author: yyangsc@mail.tsinghua.edu.cn.

little reward-driven exploration and still be difficult to classify relative to the near-optimality boundary.

We study these objectives together under permanent exogenous arrivals and stationary reward laws. If $r^*(t)$ is the largest mean among the arms available at time t, an arm is relatively near-optimal when its mean is at least $r^*(t) - \varepsilon$. Reports certify membership, certify exclusion, or leave the arm unresolved. We require one simultaneous correctness event for all labels at all times. To rule out indefinite abstention, we charge the cumulative unresolved time of arms whose means lie more than η from the membership boundary. The question is how much regret is necessary, and how much is attainable, under this reporting requirement.

New arrivals affect the task in two ways. An uncertain new candidate may be better than the incumbent by enough to invalidate many existing positive reports. Evidence about that one candidate is then relevant to many reports at once. A later improvement can also make an earlier difficult membership question unnecessary: a candidate close to the old boundary may be far below the new one. Its existing upper confidence bound can be combined with a new reference's lower bound to certify exclusion. Neither effect requires the reward law of an old arm to change.

Our observation model and regret comparator have established precedents. Permanent arrivals form a monotone special case of sleeping-bandit availability (Kleinberg et al., 2010); ballooning bandits already compare reward with the best arm currently available (Ghalme et al., 2021). Exogenous arrivals differ from drawing arms on demand from a reservoir (Berry et al., 1997) and from the disappearing actions of mortal bandits (Chakrabarti et al., 2008). Benchmark improvements here arise from new arms, whereas nonstationary bandits allow the reward distributions of existing arms to change (Besbes et al., 2014; Auer et al., 2019). Sequential allocation and adaptive experimentation have a long statistical history (Robbins, 1952; Lai and Robbins, 1985; Lai, 1987); the expanding set of actions retains that allocation problem and adds an evolving inferential target.

For a fixed arm set, identifying every relatively near-optimal arm is the ALL-ε problem (Mason et al., 2020). Joint reward and identification objectives also have established precedents (Degenne et al., 2019; Yang et al., 2026; Cho et al., 2025). Our task requires continuing reports under arrivals, with all-time label reliability and an explicit cumulative cost for unresolved arm time. The distinction concerns the reporting protocol and its cost, rather than a new arriving-arm model. The simultaneous recovery scales used in our lower bounds build on existing information inequalities and multiple-recovery arguments (Kaufmann et al., 2016; Malloy and Nowak, 2014; Jamieson and Jain, 2018).

We give three results about maintenance cost. First, a common-reference lower bound shows how one uncertain arrival can obstruct many other certifications. Its information queue accumulates cost over time within one reward environment. Second, we obtain matching worst-schedule orders for cumulative unresolved area, including finite-horizon truncation, and derive the polynomial arrival boundary for jointly sublinear area and expected regret. This worst-case lower bound already occurs in an initial static batch; the common-reference result describes a separate consequence of continuing arrivals. Third, on a specified repeated-update family, rapid resolution of temporary membership tasks has matching inverse-square regret costs. Allowing a constant-factor larger area

budget admits a logarithmic regret upper bound. A nonvanishing-probability lower bound holds in one fixed environment under the same cumulative-area requirement. The deferred endpoint has only an upper bound, and the general frontier over intermediate budgets and overlapping arrivals remains open.

An explicit fixed-precision procedure, Checkpoint-UCB, supplies the attainability results. It shares all observations between certification and reward learning and retains every arrived arm in reward selection. The construction separates the statistical costs cleanly, but its conservative quotas can exceed a practical budget. Our numerical study retains those failures as well as favorable comparisons. Section 2 defines the joint task, Section 3 gives the construction, and Sections 4–5 establish the maintenance and regret bounds. Section 6 examines their finite-budget implications.

2 Model, reliable reports, and maintenance cost

2.1 Observation model and regret

Let $\mathcal{A}$ be a possibly infinite set of arms. Arm i arrives at time τ_i, after which it remains available. At every physical time $t = 1, 2, \ldots,$

$$\mathcal{A}_t = \{i : \tau_i \le t\}, \qquad 1 \le K_t = |\mathcal{A}_t| < \infty.$$

Arbitrary finite batches are allowed, including $K_T > T$ at an evaluation endpoint. Each arm has an unknown stationary reward law on $[0, 1]$ with mean r_i. Its successive potential observations $(Y_{i,n})_{n\ge1}$ are independent and identically distributed, and the streams of different arms are independent. Arrivals are exogenous and independent of both these streams and policy randomization. Unless stated otherwise, each result is for a fixed deterministic arrival schedule.

At time t, the policy first receives all current arrivals, then selects one arm $I_t \in \mathcal{A}_t$, observes one reward X_{t,I_t}, and finally issues its report. It uses the observed history

$$H_{t-1} = (\mathcal{A}_s, I_s, X_{s,I_s})_{s=1}^{t-1}$$

and independent randomization. Means, future arrivals, and true membership labels are unavailable to the policy. With $N_i(t) = \sum_{s\le t} \mathbf{1}\{I_s = i\}$, the physical observation is $X_{t,I_t} = Y_{I_t,N_{I_t}(t-1)+1}$. Write

$$\hat{r}_i(n) = \frac{1}{n}\sum_{s=1}^{n} Y_{i,s}, \qquad r^*(t) = \max_{i\in\mathcal{A}_t} r_i, \qquad \Delta_i(t) = r^*(t) - r_i \quad (i \in \mathcal{A}_t).$$

At a deterministic endpoint T, cumulative dynamic regret is

$$R_T(\pi) = \sum_{t=1}^{T} \Delta_{I_t}(t).$$

This is a sum of mean gaps, random through the policy's arm selections; $\mathbb{E}R_T$ is its expectation. The quantity does not compare realized reward sequences. Physical time always counts actual

observations.

2.2 Reliable reports and cumulative unresolved time

For $0 < \varepsilon < 1$, the target set is

$$\mathcal{G}_t^\varepsilon = \{i \in \mathcal{A}_t : \Delta_i(t) \le \varepsilon\}.$$

After the current observation, a report partitions all of $\mathcal{A}_t$ into $\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t$: certified members, certified nonmembers, and unresolved arms. Define

$$E_{\text{lab}} = \{\text{for all } t:\ \mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t \setminus \mathcal{N}_t\}.$$

A policy is δ-reliable if $\mathbb{P}(E_{\text{lab}}) \ge 1 - \delta$ in every allowed reward environment. A positive report may be withdrawn after a new arrival. On the correctness event, a negative report remains valid because $r^*(t)$ is nondecreasing and the arm's mean is stationary.

Reliability permits all arms to remain unresolved. For a boundary width $0 < \eta < 1$, define

$$\begin{aligned} d_i^\varepsilon(t) &= |\Delta_i(t) - \varepsilon|, \\ U_t(\eta) &= \sum_{i \in \mathcal{A}_t} \mathbf{1}\{i \in \mathcal{V}_t, d_i^\varepsilon(t) > \eta\}, \qquad \mathcal{U}_T(\eta) = \sum_{t=1}^{T} U_t(\eta). \end{aligned} \tag{1}$$

The count includes newly arrived arms with no observations and old arms that become unresolved again. Every such arm–time pair is charged; the loss is not limited to the time before a first decision. Within the boundary band, unresolved time is exempt, but any label issued there must still be correct. The true distance $d_i^\varepsilon(t)$ is used only by the evaluator and in the analysis.

The task parameters are fixed when comparing policies. Their roles are summarized below.

Parameter	Meaning
$\varepsilon \in (0,1)$	Relative near-optimality tolerance
$\delta \in (0, 1/2)$	Joint error budget
$\eta \in (0,1)$	Boundary width exempting unresolved time
$\rho \in (0,1)$	Predetermined precision of reporting checkpoints

The main requirement at a fixed T is

$$\mathbb{E}_\theta R_T \le r, \qquad \mathbb{P}_\theta\{E_{\text{lab}}, \mathcal{U}_T(\eta) \le a\} \ge 1 - \delta. \tag{2}$$

It combines an expected regret bound with a joint high-probability label–area guarantee. In particular, $\mathcal{U}_T = o(T)$ requires the time-average number of unresolved separated arms to vanish, rather than only a small unresolved fraction of all arms. The area guarantee is for each deterministic endpoint; it need not hold for every endpoint on one common area event.

For a performance family Θ, let $\mathfrak{R}_\Theta(a)$ be the infimum of $\sup_{\theta \in \Theta} \mathbb{E}_\theta R_T$ over policies reliable on the full bounded model and satisfying the probability condition in (2) for every $\theta \in \Theta$. The value is

$+\infty$ if the feasible class is empty. Full-model label reliability and a family-specific area guarantee are distinct requirements. The lower bounds remain valid when reliability is required only on the lower-bound subfamily.

2.3 How arrivals change the reporting problem

For a fixed old arm, $\Delta_i(t)$ is nondecreasing, so its true membership can change only from member to nonmember. Its boundary distance $|\Delta_i(t) - \varepsilon|$ need not be monotone. A new candidate can first create uncertainty about the current best mean, obstructing positive certification of many old arms even when those arms remain truly near-optimal. A sufficiently good new reference can later move the boundary past earlier candidates, making their fine classification unnecessary. Sections 4 and 5 quantify these two effects under the same reporting protocol.

3 A checkpoint construction for joint guarantees

The upper bounds use a fixed-precision construction, Checkpoint-UCB. Its reporting intervals are formed once, while its reward estimates use every observation. Index arms by arrival order $i = 1, 2, \ldots$, with a fixed order within each batch. Choose $0 < \rho < 1$ before observing rewards and set

$$q_i(\rho) = \left\lceil \frac{2}{\rho^2} \log \frac{4i(i+1)}{\delta} \right\rceil. \tag{3}$$

Before this lifetime count is reached, arm i has reporting interval $[0, 1]$. At its q_ith observation the interval becomes permanently

$$[\ell_i, u_i] = [\hat{r}_i(q_i) - \rho/2, \hat{r}_i(q_i) + \rho/2] \cap [0, 1]. \tag{4}$$

Although its completion time is adaptive, the interval uses a predetermined potential prefix. Different precisions ρ_i may also be prescribed before the corresponding arms are observed; data-dependent checkpoint selection is not covered.

At even times the policy serves the first incomplete checkpoint in arrival order, if one remains. At every other time it maximizes, over all $i \in \mathcal{A}_t$,

$$J_i(t) = \hat{r}_i(N_i(t-1)) + \sqrt{\frac{3\log(2t)}{N_i(t-1)}}. \tag{5}$$

A zero count gives index $+\infty$; ties are broken by the smallest arrival-order label. Both kinds of pull update the same count and reward sum. After observing the selected reward, define $\underline{r}^*(t) = \max_{i \in \mathcal{A}_t} \ell_i$ and $\overline{r}^*(t) = \max_{i \in \mathcal{A}_t} u_i$, and report

$$\mathcal{P}_t = \{i : \ell_i \geq \overline{r}^*(t) - \varepsilon\}, \qquad \mathcal{N}_t = \{i : u_i < \underline{r}^*(t) - \varepsilon\}, \qquad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t). \tag{6}$$

Algorithm 1 observes exactly one reward per physical round. Its variables n_i and s_i are the lifetime count and reward sum. An ordinary UCB pull can complete any checkpoint, including one

Algorithm 1 Checkpoint-UCB

Require: $\varepsilon \in (0,1)$, $\delta \in (0,1/2)$, predetermined $\rho \in (0,1)$

$\mathcal{A}^{\mathrm{cur}} \leftarrow \varnothing$, FIFO queue $\mathcal{Q}^{\mathrm{srv}} \leftarrow \varnothing$, $k \leftarrow 0$
for physical time $t = 1, 2, \ldots$ **do**
  for each arm arriving at t, in a fixed within-batch order **do**
    $k \leftarrow k+1$; assign label $i \leftarrow k$; $\mathcal{A}^{\mathrm{cur}} \leftarrow \mathcal{A}^{\mathrm{cur}} \cup \{i\}$
    $n_i \leftarrow 0$, $s_i \leftarrow 0$, $[\ell_i, u_i] \leftarrow [0,1]$
    $q_i \leftarrow \lceil (2/\rho^2) \log\{4i(i+1)/\delta\} \rceil$; append i to $\mathcal{Q}^{\mathrm{srv}}$
  end for
  while $\mathcal{Q}^{\mathrm{srv}} \neq \varnothing$ and $n_{\mathrm{front}(\mathcal{Q}^{\mathrm{srv}})} \geq q_{\mathrm{front}(\mathcal{Q}^{\mathrm{srv}})}$ **do**
    Remove the front of $\mathcal{Q}^{\mathrm{srv}}$
  end while
  if t is even and $\mathcal{Q}^{\mathrm{srv}} \neq \varnothing$ **then**
    $I \leftarrow \mathrm{front}(\mathcal{Q}^{\mathrm{srv}})$ ▷ designated service pull
  else
    $J_i \leftarrow +\infty$ if $n_i = 0$, and $J_i \leftarrow s_i/n_i + \sqrt{3\log(2t)/n_i}$ otherwise, for $i \in \mathcal{A}^{\mathrm{cur}}$
    $I \leftarrow \min \arg\max_{i \in \mathcal{A}^{\mathrm{cur}}} J_i$
  end if
  Pull I once and observe x; $n_I \leftarrow n_I + 1$, $s_I \leftarrow s_I + x$
  if $n_I = q_I$ **then**
    $m \leftarrow s_I/q_I$; $\ell_I \leftarrow \max\{0, m - \rho/2\}$, $u_I \leftarrow \min\{1, m + \rho/2\}$
  end if
  $L \leftarrow \max_{i \in \mathcal{A}^{\mathrm{cur}}} \ell_i$, $U \leftarrow \max_{i \in \mathcal{A}^{\mathrm{cur}}} u_i$
  $\mathcal{P}_t \leftarrow \{i \in \mathcal{A}^{\mathrm{cur}} : \ell_i \geq U - \varepsilon\}$, $\mathcal{N}_t \leftarrow \{i \in \mathcal{A}^{\mathrm{cur}} : u_i < L - \varepsilon\}$
  $\mathcal{V}_t \leftarrow \mathcal{A}^{\mathrm{cur}} \setminus (\mathcal{P}_t \cup \mathcal{N}_t)$; report $(\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t)$
end for

behind the queue's front. The queue discards such completed entries when they reach its front. An interval is computed only when $n_i = q_i$, and is not recomputed when $n_i > q_i$. Reaching q_i forms fixed-precision evidence; it need not produce a determinate report if other arms are still uncertain or the arm lies closer to the boundary than that evidence can resolve.

All arrived arms remain eligible for (5), including negatively reported arms. The intervals are nonempty on every path, so $\mathcal{P}_t$ and $\mathcal{N}_t$ are disjoint. A count, sum, and interval are stored for every arm; a direct index or report update costs $O(K_t)$ operations. The policy uses no terminal horizon, future arrivals, true means, or reference identities.

Proposition 3.1 (Simultaneous reporting and unconditional reward cost)**.** *On the full bounded model, Checkpoint-UCB satisfies $\mathbb{P}(E_{\mathrm{lab}}) \geq 1 - \delta/2$. If $\mathcal{I}_T$ is its set of designated service time steps, then at every deterministic T,*

$$\mathbb{E}R_T \leq \min\left\{T, K_T + 4\sqrt{3K_T T \log(2T)} + 1 + \mathbb{E}\sum_{t\in\mathcal{I}_T} \Delta_{I_t}(t)\right\}. \tag{7}$$

Pathwise, $|\mathcal{I}_T| \leq \sum_{i\leq K_T} q_i$.

Hoeffding's inequality bounds the failure at checkpoint i by $\delta/\{2i(i+1)\}$. Summing over potential arm indices gives one coverage event of probability at least $1 - \delta/2$, on which every report is correct. For (7), a separate, summable concentration argument at physical time t controls ordinary UCB pulls. Each arm and pre-pull count is charged at most once; inserted service observations only skip counts. This expectation bound holds without conditioning on reporting success, so it omits no δT term. It bounds one policy's total regret, rather than a regret difference between its trajectory and a hypothetical policy without service. Complete proofs appear in the supplementary material.

4 Fundamental costs of continual maintenance

4.1 One uncertain arrival can obstruct many reports

The first lower bound retains one reward environment throughout the time axis. It therefore measures a cumulative burden without adding the worst cases of incompatible experiments. Fix $0 < \varepsilon \leq 1/16$, $0 < \delta \leq 1/16$, and $0 < \eta < \varepsilon/2$. An initial reference has Bernoulli mean $1/4+\varepsilon/2$. Candidates arrive on any prescribed schedule. In environment zero each has mean $p = 1/4$; environment j changes only candidate j to mean $p + 2\varepsilon$. All candidates are good in environment zero, with boundary distance $\varepsilon/2$. After candidate j arrives in environment j, every other candidate is bad.

Let M_t count arrived candidates and $a_t = M_t - M_{t-1}$. Define

$$q_- = \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \qquad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0,$$
$$\mathcal{B}_T = \sum_{t\leq T} (M_t - 1)_+ \mathbf{1}\{Q_t > 0\}. \tag{8}$$

This analytical queue is an information-capacity certificate: arrivals add q_- units per candidate,

while only one observation is possible per round. It is distinct from the algorithm's FIFO queue $\mathcal{Q}^{\mathrm{srv}}$ and does not impose a pathwise sampling quota on every candidate. In the stated range $q_- \geq 2$.

Theorem 4.1 (Broadcast uncertainty)**.** *For every policy reliable on environment zero and all its one-candidate alternatives,*

$$\mathbb{E}_0 \mathcal{U}_T(\eta) \geq (3/4 - \delta)\mathcal{B}_T.$$

If $\mathcal{B}_T > 0$*, then*

$$\mathbb{P}_0\{\mathcal{U}_T(\eta) \geq \mathcal{B}_T/4\} \geq 1/4.$$

These conclusions still hold when the policy knows the full arrival schedule, the horizon, and the reference distribution.

At a busy time, some deterministic candidate label has expected sample count below q_-. Changing that candidate alone has too little history KL to permit positive certification of another candidate with probability at least $1/4$. Reliability also bounds incorrect negative reports. The remaining probability must be unresolved. The same reference environment supports this argument at every time; the difficult label is selected from expected counts rather than from a favorable realized history. The supplementary proof includes the bounded-variable step needed for the probability statement.

If m candidates have arrived by time s, the queue obeys the deterministic inequality

$$\sum_{t \leq s+qm-1} (M_t - 1)_+ \mathbf{1}\{Q_t(q) > 0\} \geq \frac{q-1}{2} m(m-1), \qquad q \geq 2. \tag{9}$$

The factor m^2 reflects the number of affected reports as well as the work needed to exclude high-quality alternatives. The subtraction of one unit in $q - 1$ accounts for the last observation of each busy period: reports are issued after service.

4.2 Worst-schedule maintenance area

Let $\mathfrak{A}_{T,K}$ be the smallest a attainable by a full-model reliable policy with

$$\sup_{\boldsymbol{\tau}: K_T \leq K} \sup_{\boldsymbol{r}} \mathbb{P}_{\boldsymbol{r}}\{E_{\mathrm{lab}}^c \text{ or } \mathcal{U}_T(\eta) > a\} \leq \delta.$$

This optimization imposes no regret budget. Policies may know T, K and the schedule for the lower bound. The upper bound uses neither T nor K as input.

Theorem 4.2 (Worst-schedule area and joint attainability)**.** *For integers* $K \geq 2$ *and* $T \geq 1$*,* $0 < \varepsilon \leq 1/8$*,* $0 < \eta \leq \varepsilon/32$*, and* $0 < \delta \leq 1/16$*,*

$$2^{-24} \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\} \leq \mathfrak{A}_{T,K} \leq 256 \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\}. \tag{10}$$

Checkpoint-UCB with $\rho = \eta/4$ attains the upper order and also satisfies

$$\mathbb{E}R_T \le \min\left\{T, K + 4\sqrt{3KT\log(2T)} + 1 + 99K\eta^{-2}\log(K/\delta)\right\}. \tag{11}$$

For the upper bound, completed checkpoints imply endpoint errors at most $\eta/4$, sufficient to classify all arms outside the boundary band. The unfinished-work queue is dominated by adding q_i at each arrival and serving at even rounds. On the simultaneous coverage event,

$$\mathcal{U}_T(\eta) \le \min\left\{K_T T, 2K_T \sum_{i \le K_T} q_i\right\}. \tag{12}$$

Additional UCB observations reduce the work; they are not charged again as new checkpoint samples. The lower bound uses $K-1$ candidates with independent means $1/2 \pm 2\eta$ and a reference of mean $1/2 + \varepsilon$, all available at the first pull. Simultaneous partial recovery and a horizon truncation give the probability statement required by the definition of $\mathfrak{A}_{T,K}$. The statistical $\log K$ factor has fixed-arm multiple-testing precedents; this worst-schedule lower construction itself requires no repeated arrivals.

4.3 Growth rates and the limits of total-count criteria

Corollary 4.3 (Arrival-growth boundary)**.** *Fix the task parameters of Theorem 4.1 and one infinite arrival schedule. If a policy is reliable for every reward environment and has a deterministic envelope $a_T = o(T)$ satisfying (2) in every environment, then $K_t = o(\sqrt{t})$. Conversely, $K_T^2 \log(eK_T) = o(T)$ suffices for one horizon-free policy to have both $\mathbb{E}R_T = o(T)$ and such a sublinear area envelope. In particular, for $K_t \asymp t^\alpha$, $0 < \alpha < 1$, simultaneous attainment is possible if and only if $\alpha < 1/2$.*

The necessity follows from (9). For sufficiency, the general checkpoint completion bound in Supplementary Lemma S2.1 applies throughout $0 < \eta < \varepsilon/2$, together with Proposition 3.1. This argument does not require the narrower range $\eta \le \varepsilon/32$ in Theorem 4.2. The exponent statement leaves a logarithmic gap for general critical schedules.

Within the parameter range of Theorem 4.2, (10) also gives a uniform terminal-count statement: over all schedules with $K_T \le k(T)$, sublinear expected regret and sublinear service envelopes are jointly attainable if and only if $k(T)^2 \log k(T) = o(T)$. Here the worst schedule can change with T. This statement concerns a different schedule class from the single infinite schedule in Corollary 4.3.

The necessary condition $K_t = o(\sqrt{t})$ is not sufficient on every fixed schedule. For example, batches of size

$$m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor, \qquad s_n = 2^{2^n},$$

give $K_t = o(\sqrt{t})$ but preclude any $o(T)$ area envelope uniform over reward environments. Near s_n, simultaneous classification of that burst requires accumulated unresolved area of order $m_n^2 \log m_n$, which exceeds s_n by a growing factor. The supplementary argument selects a single infinite schedule; the difficult reward environment is allowed to vary with the endpoint, as required by the stated

uniform guarantee.

5 Temporary membership tasks and regret trade-offs

5.1 A repeated-update family

We keep the reward environment, arrival schedule, endpoint, and reporting parameters fixed when comparing two area budgets. Choose integers $J \geq 1$, $M \geq \max\{2, J\}$ and

$$0 < \varepsilon \leq 1/8, \quad 0 < \delta \leq 1/16, \quad 0 < v, \quad Jv \leq \varepsilon/4, \quad 0 < d \leq v/32, \quad \eta = d/2. \tag{13}$$

At $a_e = 1 + eH$, $e = 0, \ldots, J$, a reference with mean $b_e = 1/2 + ev$ arrives. For each $e < J$, the same batch contains M candidates with means

$$r_{e,k} = b_e - \varepsilon + \sigma_{e,k} d, \qquad \sigma_{e,k} \in \{-1, +1\}. \tag{14}$$

All rewards are Bernoulli and all arms remain available. The last batch contains only its reference, and $T = (J + 1)H$. The family Θ includes every sign configuration and every within-batch permutation. The policy is not told which arm is the reference. Lower bounds may disclose those identities and the schedule.

In its birth window a candidate is good exactly when its sign is positive. Its reward gap is $\varepsilon - \sigma_{e,k} d$, while its distance to the membership boundary is d. At the next update its gap becomes $\varepsilon + v - \sigma_{e,k} d$, so both signs are bad, with boundary distance at least $v - d$. Every reference remains good: the final good set consists of all references. The tasks that distinguish the candidate signs expire even though their arms remain available. The restriction $Jv \leq \varepsilon/4$ limits the number of updates at any fixed positive improvement size.

5.2 Low area forces enough temporary tasks to finish

The lower-bound ingredient permits arbitrary interleaving and stopping. For m independent uniform signs with observation laws P_i^+, P_i^-, put

$$\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty).$$

If a bounded stopping time has at least half the signs determined with prior probability at least $1/2$, and all configurations have familywise report error at most $\delta \leq 1/16$, then

$$\overline{\mathbb{E}} \sum_{i=1}^{m} N_i \geq \frac{m}{1024\kappa} \log \frac{m}{\delta}. \tag{15}$$

Other observations must not encode these signs. The proof factors the conditional posterior after the adaptive sampling and bounded stopping kernels cancel, and combines simultaneous recovery with coordinatewise change of measure (Kaufmann et al., 2016). It does not assume independence

of the random sample counts. The product prior is a proof device; the final lower bound concerns a fixed environment.

A window that never reaches half-completion contributes at least $MH/2$ area. Thus, on the event $\mathcal{U}_T(\eta) \le V/8$, where $V = JMH$, at least $3J/4$ windows half-complete. A common product prior over all windows turns this area requirement into enough instances of (15). Each candidate observation costs at least $\varepsilon - d$ regret, and observations from distinct batches are charged once. This argument does not impose a deadline on every individual batch; the policy can choose where to defer. The heterogeneous completion probabilities and their information-allocation relaxation are given in Supplementary Section S6. That relaxation is a necessary constraint, with no general causal attainability claim.

5.3 Shared evidence for area and reward guarantees

Checkpoint-UCB may use a predetermined precision ρ_e for each arriving batch. Consider the heterogeneous repeated-update family: batch F_e has reference mean b_e and M_e candidate means $b_e - \varepsilon \pm d_e$, all in the allowed reward range. The windows partition physical times $1, \dots, T$; write a_e for the arrival time of batch e, H_e for its window length, and $S_e = \sum_{i \in F_e} q_i(\rho_e)$ for its total checkpoint work; the final batch contains only a reference. Set $M_{-1} = M_J = 0$. Suppose

$$H_e \ge 2S_e, \qquad b_J - b_0 \le \varepsilon/4, \qquad d_e + \rho_e + \rho_{e+1} < b_{e+1} - b_e, \qquad 2\max_e \rho_e \le \varepsilon/2. \tag{16}$$

The neighboring-batch condition is imposed for $e < J$; the window and precision bounds include the final batch. These are conditions on the performance family, not future information supplied to the policy. Put $\mathcal{F} = \{e < J : 2\rho_e < d_e\}$. For any arm that has a positive gap at some time through T, define

$$\gamma_i = \min_{t \le T : i \in \mathcal{A}_t,\, \Delta_i(t) > 0} \Delta_i(t), \qquad D_i^T = \max_{t \le T : i \in \mathcal{A}_t} \Delta_i(t).$$

Only such arms are included in the sum containing γ_i below.

Proposition 5.1 (Shared update from lifetime evidence)**.** *Under (16), on the checkpoint coverage event,*

$$\mathcal{U}_T(\eta) \le \sum_{e \notin \mathcal{F}} M_e H_e + 2\sum_{e \in \mathcal{F}} M_e S_e + 2\sum_{e=0}^{J} (e + 1 + M_{e-1}) S_e. \tag{17}$$

When the next batch completes its checkpoints, every preceding-batch candidate can be excluded using its saved interval and the new reference lower bound, without another observation of that old candidate for certification. For the same policy, the unconditional expected regret satisfies

$$\begin{aligned} \mathbb{E}R_T \le & \sum_{e<J} \sum_{k=1}^{M_e} (\varepsilon + d_e) q_{e,k} \\ & + \sum_{i:\gamma_i \text{ exists}} D_i^T \left[\left\lceil \frac{12\log(2T)}{\gamma_i^2} \right\rceil - q_i + 1 \right]_+ + 1. \end{aligned} \tag{18}$$

The last area term charges temporarily affected references and the preceding candidate cohort, rather than all historical candidates at every update. For $d_e > \eta$, batches in $\mathcal{F}$ complete their current fine membership task; a batch with $d_e \le \eta$ lies inside the evaluation band in its birth window, and its corresponding area term is conservative. An old candidate's upper endpoint and the new reference's lower endpoint differ by more than ε, giving the later negative report without new observations of that candidate. This does not prevent the reward layer from selecting it again.

Every q_i is a lifetime quota. The condition $H_e \ge 2S_e$ ensures that all observations up to that quota occur in the arm's birth window. The first term of (18) charges candidate observations in that window, while reference observations then have zero regret. The second term counts excess ordinary UCB pulls using distinct pre-pull counts. The expectation is not conditioned on checkpoint coverage. The birth-window condition is essential to this charging and does not cover arbitrary overlapping arrivals.

5.4 Rapid and deferred maintenance at the same endpoint

For the homogeneous family (13)–(14), set

$$K = J(M+1)+1, \quad L_K = \log\frac{4K(K+1)}{\delta}, \quad \rho_f = d/4, \quad \rho_c = v/8,$$
$$q_f = \lceil 32L_K/d^2 \rceil, \quad q_c = \lceil 128L_K/v^2 \rceil, \quad H = 128(M+J+1)q_f, \quad V = JMH, \quad T = (J+1)H. \tag{19}$$

The quantities q_f, q_c are uniform analysis bounds on the actual armwise quotas. Neither final arm count nor future times are policy inputs. Fine precision can be set from the task input as $\rho_f = \eta/2$; the coarse precision is a separate preselected configuration. Automatic adaptation to an unknown improvement scale is not asserted.

Theorem 5.2 (Rapid and deferred maintenance at the same endpoint)**.** *On the preceding family and horizon,*

$$2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \le \mathfrak{R}_\Theta(V/8) \le 2^{18}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}. \tag{20}$$

The coarse-precision Checkpoint-UCB policy has joint label-and-area success probability at least $1-\delta$ with area budget $9V/8$, and

$$\sup_{\theta\in\Theta}\mathbb{E}_\theta R_T \le 2\varepsilon JMq_c + 64\left(\frac{JM}{\varepsilon}+\frac{J^2}{v}\right)\log(2T)+1. \tag{21}$$

Consequently, at fixed legal $J, M, \varepsilon, v, \delta$ and $d \downarrow 0$, with $\eta = d/2$ and the horizon in (19), $\mathfrak{R}_\Theta(V/8) = \Theta(d^{-2})$, whereas $\mathfrak{R}_\Theta(9V/8) = O(\log(1/d))$.

For the lower bound, low area forces enough windows to satisfy (15). For the upper bound, (17) gives the two area budgets. At the specified horizon the deterministic bound $R_T \le 2\varepsilon T$ suffices for the rapid order; it would not justify the same conclusion at arbitrarily longer windows. The coarse bound instead counts checkpoint observations and excess ordinary UCB pulls separately, as in (18).

The two budgets concern the same $\Theta, \varepsilon, \eta, \delta, T$, but allow different service levels. Their ratio does not describe an individual delay for every arm. The theorem does not claim that the coarse policy meets the rapid budget, that its regret upper bound is matched, or that all intermediate budgets are characterized. A preselected mixture of fine and coarse batches gives additional attainable points in the supplement. The separation concerns this specified repeated-update family, with η and T changing together as $d \downarrow 0$.

5.5 A probability lower bound under the same area requirement

An expectation lower bound alone could reflect unusual sample paths. The following result gives a stronger conclusion without replacing cumulative area by individual deadlines.

Theorem 5.3 (A nonvanishing probability of the rapid-maintenance cost)**.** *In the repeated-update family, suppose a reliable policy satisfies*

$$\mathbb{P}_\sigma\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le V/8\} \ge 1 - \delta$$

for every sign configuration. Then one fixed configuration satisfies

$$\mathbb{P}_\sigma \left\{ R_T \ge 2^{-20} \frac{\varepsilon J M}{d^2} \log \frac{M}{\delta} \right\} \ge \frac{11}{80}. \tag{22}$$

The proof stops each window at half-completion, its endpoint, or its $(q+1)$st candidate observation, where q is a fixed sample budget. Stopping at the first observation beyond q retains reports that become determinate after a reference observation while the candidate count is still q. Applying the partial-recovery lemma to this bounded experiment makes the probability of a cheap completion less than $1/2$. Low total area requires many completions; counting cheap ones then leaves enough expensive windows with probability at least $1/5 - \delta \ge 11/80$. Their actual observations give (22). No independence between window-completion events is needed. Averaging under the common prior then selects one fixed environment.

Since $11/80 > \delta$, the rapid endpoint in (20) also has matching order when regret and area must be bounded on one event of probability at least $1 - \delta$. The exact joint-quantile definition and corollary are in Supplementary Section S8. This consequence concerns the rapid endpoint at the prescribed horizon and gives no matching joint-quantile result for deferred maintenance.

6 Numerical studies

The experiments compare reward allocation and reliable reporting at a fixed physical budget. They ask how the service rule and the precision of its reporting evidence affect regret and unresolved area. The four environments are not the restricted family used for the two-endpoint separation result, so they do not test its asymptotic rates.

6.1 Protocol and competing rules

All rewards are Bernoulli. We fix $T = 20{,}000$, $\varepsilon = .30$, $\eta = .10$, and $\delta = .05$, with eight arms by the end of each run. There are 160 independent replications per environment. Within a replication, each method receives the same potential reward stream for each arm: its nth pull of arm i reveals the same $Y_{i,n}$. The shared streams do not force the methods to take the same actions. Arrivals precede the action, and the report follows the observation. All methods retain previous observations. The configuration records the arrival times, means, seeds, and policy parameters; a policy receives current arrivals and observed history, without future arrivals, true means, or the evaluation horizon.

The stable-benchmark environment starts with means $.65, .50$ and adds three batches of inferior arms. The successive-improvement environment has benchmark levels $.45, .70, .90, .98$, so a later arrival can change an unresolved membership question. The burst environment introduces six arms at one time, including a new best arm. In the fourth environment arrivals stop at pull 4037, leaving nearly 16,000 pulls for recovery. Noninitial batches arrive 36 pulls after a 100-pull admission grid; the short-period baseline is therefore not evaluated only at its most favorable alignment.

We use two FIFO configurations of Checkpoint-UCB, denoted CP-FIFO in the tables and CP in the figures, with service every second pull. The finer choice $\rho = .04$ satisfies $2\rho < \eta$. Once every currently available arm has completed its checkpoint, all η-separated arms are resolved on the coverage event. This is not the $\rho = \eta/4 = .025$ calibration used for the worst-case area bound. The coarser choice $\rho = .12$ retains reliable labels but does not guarantee completeness for arbitrary environments at $\eta = .10$. Their eight-arm quota totals are 73,083 and 8,123, respectively. In particular, the finer configuration cannot complete all quotas within the physical budget.

Three further rules retain the coarser checkpoint report and change allocation: CP-period8 serves the FIFO queue every eighth pull; CP-confidence uses confidence-driven service every second pull; and UCB-checkpoint uses full-history UCB without designated service. The confidence-driven rule samples the widest interval among unresolved arms and the arm with largest upper endpoint, breaking ties by sample count and arrival order. It stops designated service when every report is resolved, and a completed checkpoint keeps its fixed interval. These are separately identified configurations; the FIFO guarantee is not automatically a guarantee for every allocation variant.

Three methods combine the full-prefix confidence report with a reward or sampling rule. UCB-prefix admits every arm immediately and uses the same full-history UCB index as Checkpoint-UCB. Growing-MOSS uses

$$\hat{r}_i(n) + \sqrt{\frac{\max\{\log(t/(K_t n)), 0\}}{n}},$$

with infinite index at $n = 0$. We do not transfer the fixed-horizon MOSS minimax theorem to this growing-arm rule. ALL-ε adaptation uses confidence-driven service every second pull and UCB otherwise. It retains history and can revise positive reports after arrivals. It is a natural adaptation of confidence-based all-good-arm sampling, rather than an implementation of FAREAST or $(\mathrm{ST})^2$.

Batch100-UCB admits waiting arms at pulls $1, 101, 201, \ldots$, applies UCB to admitted reward-active arms, and retires an arm from reward sampling when its upper bound falls below the largest active lower bound. Observations are retained. Its report still covers every physically arrived arm;

Table 1: Final cumulative dynamic regret and cumulative η-unresolved arm-time at $T = 20{,}000$. Entries are means (Monte Carlo standard errors), from 160 paired replications. Area is in units of 10^3. The two checkpoint precisions have different completeness guarantees.

Scenario	Method	Regret	Area / 10^3
Stable benchmark	CP-FIFO ($\rho = .04$)	1360.5 (1.5)	111.78 (0.00)
	CP-FIFO ($\rho = .12$)	3629.9 (0.7)	49.72 (0.23)
	UCB-prefix	441.7 (2.3)	83.97 (0.46)
	Growing-MOSS	77.3 (0.8)	92.43 (0.00)
	ALL-ε adapt.	1367.8 (11.6)	11.31 (0.09)
	Batch100-UCB	209.3 (2.6)	85.27 (0.52)
Successive improvements	CP-FIFO ($\rho = .04$)	5297.4 (1.2)	74.82 (0.00)
	CP-FIFO ($\rho = .12$)	1194.7 (0.0)	14.74 (0.10)
	UCB-prefix	246.2 (1.7)	18.45 (0.14)
	Growing-MOSS	53.1 (0.8)	33.65 (0.11)
	ALL-ε adapt.	2905.8 (2.4)	9.53 (0.13)
	Batch100-UCB	238.8 (2.1)	22.85 (0.09)
Burst	CP-FIFO ($\rho = .04$)	2798.0 (1.6)	75.89 (0.00)
	CP-FIFO ($\rho = .12$)	3392.2 (0.2)	28.21 (0.17)
	UCB-prefix	354.1 (2.2)	41.75 (0.39)
	Growing-MOSS	69.3 (0.9)	57.03 (0.01)
	ALL-ε adapt.	2017.4 (9.8)	10.31 (0.09)
	Batch100-UCB	278.9 (2.7)	55.87 (0.15)
Arrivals stop	CP-FIFO ($\rho = .04$)	4759.0 (1.3)	89.86 (0.00)
	CP-FIFO ($\rho = .12$)	2645.9 (0.3)	25.14 (0.01)
	UCB-prefix	248.8 (1.5)	45.96 (0.56)
	Growing-MOSS	52.0 (0.6)	71.59 (0.01)
	ALL-ε adapt.	3027.5 (4.3)	12.07 (0.07)
	Batch100-UCB	214.2 (1.8)	55.28 (0.21)

those awaiting admission have interval $[0, 1]$. Statistical exclusion, admission, and reward retirement therefore have different meanings in this comparison.

Table 1 displays six methods with means and Monte Carlo standard errors. Figure 1 includes all nine methods, with error bars of 1.96 Monte Carlo standard errors. The complete allocation table, additional paths, configuration mapping, and regret accounting are in Appendix S13. The run records contain paired differences, unresolved paths on the saved time grid, service pulls, and the regret charged to service.

6.2 Finite-budget findings and their limits

In all 640 finer-checkpoint runs, every available arm remains unresolved at every time step. Five to eight arms contribute to the final η-unresolved count because the metric exempts the boundary band; the remaining arms have not been classified. Its zero-error record therefore does not establish useful classification at this budget. The charged unresolved area ranges from 74,820 to 111,784. This is a limitation of the conservative construction at the chosen budget. The confidence-based adaptation resolves every η-separated arm by the endpoint in all replications of these four environments.

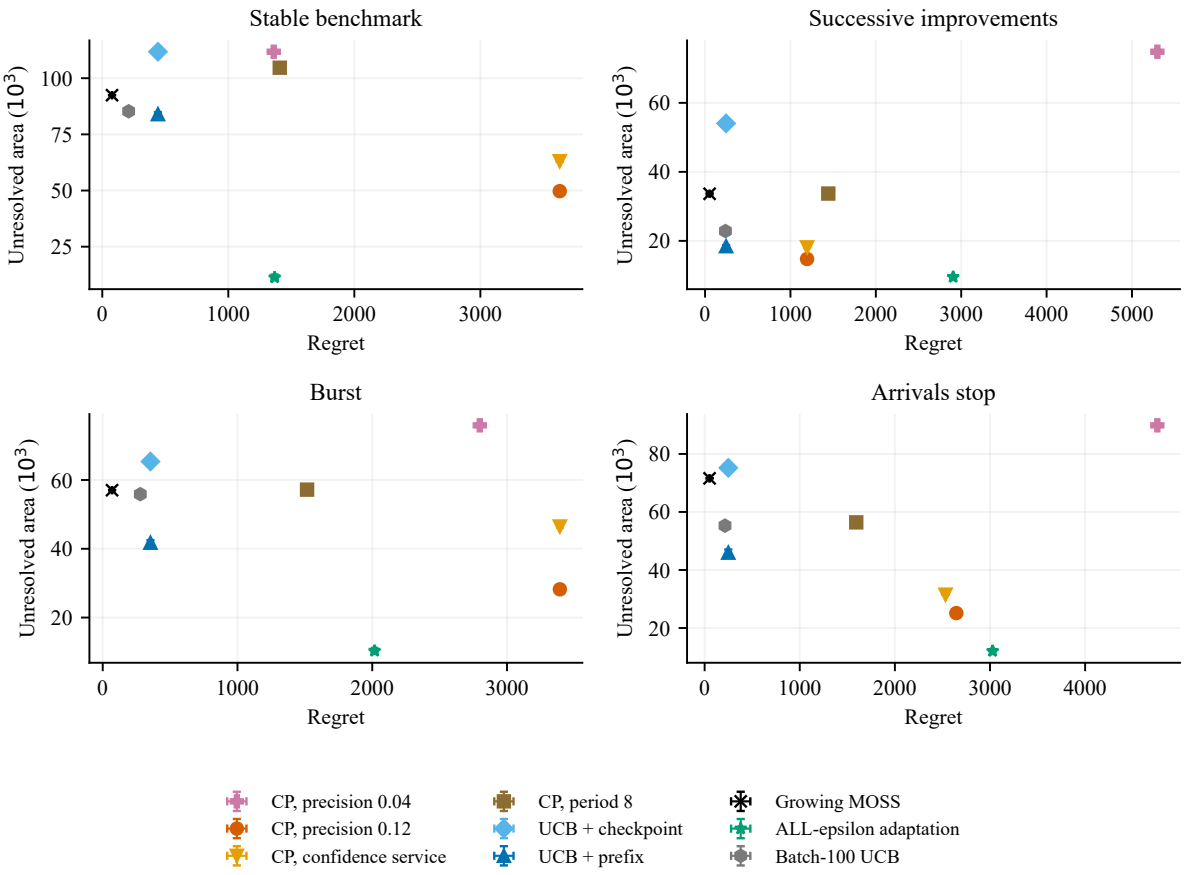


Figure 1: Regret and cumulative η-unresolved area for all nine methods. Lower values on both axes are preferable. CP denotes the Checkpoint-UCB configurations described in the text. Error bars quantify Monte Carlo uncertainty. Neither checkpoint precision nor its allocation variants uniformly dominate the confidence-based adaptation.

The coarser checkpoints reduce area relative to UCB alone, but can incur much larger regret. Under the stable benchmark, CP-FIFO has mean regret 3629.9 and area 49,718, compared with 1367.8 and 11,306 for ALL-ε adaptation. The paired differences, CP-FIFO minus the adaptation, are 2262.1 (standard error 11.7) for regret and 38,411.7 (standard error 241.9) for area. The adaptation improves both quantities in the burst environment as well. These findings do not support a general numerical advantage of checkpoint FIFO over a strong history-retaining identification rule.

Under successive improvements, CP-FIFO occupies a different cost point: regret 1194.7 and area 14,743, compared with 2905.8 and 9525 for the adaptation. The paired regret difference is -1711.0 (standard error 2.4), and the area difference is 5217.7 (standard error 141.1). This comparison does not isolate task expiration. The adaptation uses all 10,000 designated service pulls, whereas coarse CP uses about 5,786; some remaining questions fall inside the unpenalized boundary band. The regret difference can therefore partly reflect different stopping rules for reporting work. Identifying the mechanism requires a comparison with the same boundary tolerance and service requirement. These observed cost points do not estimate an optimal frontier.

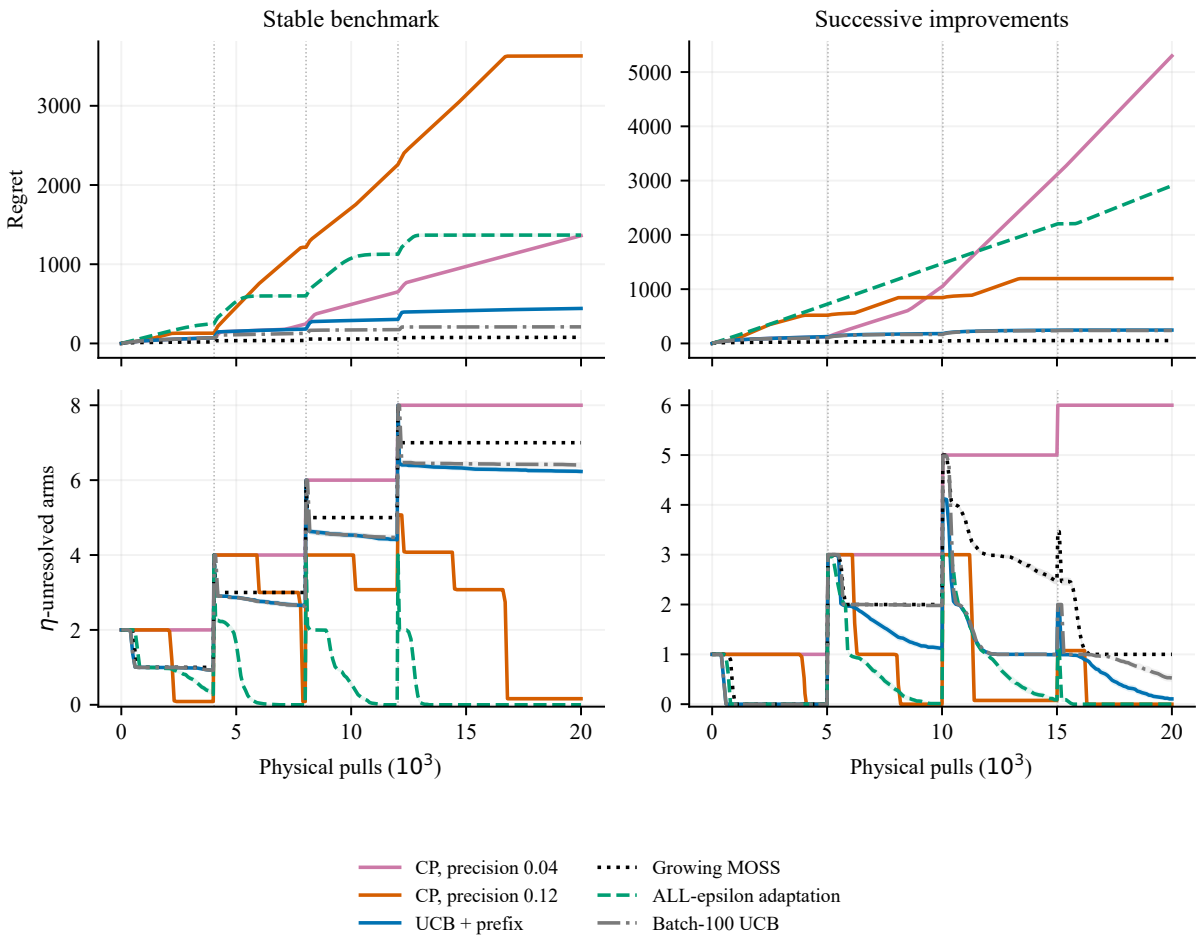


Figure 2: Mean regret and mean η-unresolved arm counts at the physical sampling clock. Bands are 1.96 Monte Carlo standard errors; vertical dotted lines mark arrivals. Stable and improving benchmarks are shown here, with the other two environments in Appendix S13. The six main-table methods are displayed. The saved results retain all nine methods and their paths on the common observation grid.

Growing-MOSS has the smallest mean regret in all four environments, between 52.0 and 77.3, while leaving substantially larger unresolved areas. Small regret alone provides little evidence of timely set maintenance; it does not rule out adding an effective certification layer to that allocation rule. Explicit service pays for observations of inferior arms. Reducing service from every second to every eighth pull generally reduces regret and enlarges unresolved area. Confidence-driven checkpoint service does not uniformly improve either criterion.

For Batch100-UCB, the admission-delay contributions to regret are 0, 33.92, 16.00, and 22.40 in the four environments. No historical retirement deficit occurred in these runs. Appendix S13 defines this decomposition and distinguishes it from learning within the reward-active set.

There were no incorrect positive or negative reports in any of the 5,760 runs. For one fixed method and environment, zero failures in 160 replications gives an exact one-sided 95% upper

confidence bound below 0.0186 on its failure probability. This finite experiment does not establish uniform reliability over all environments and times. Nor do the eight-arm comparisons validate the asymptotic lower bounds, settle the critical logarithmic gap, or establish checkpoint optimality. They quantify the finite-budget cost of the conservative quotas and show why reward and maintenance must be reported together.

7 Discussion

Reliable maintenance has costs that reward allocation alone does not measure. An uncertain arrival can obstruct many positive reports, while a later improvement can end an earlier fine classification task. The common-reference lower bound and the repeated-update construction isolate these effects. The matching worst-schedule area bound answers a different question: its lower bound is already attained by a static initial batch. The corresponding growth consequences depend on whether the guarantee concerns one infinite schedule or a uniform terminal-count class.

Checkpoint-UCB is an explicit attainability construction with a predetermined reporting precision. It reuses accumulated evidence and keeps all arrived arms eligible for rewards. Its upper bounds describe achievable statistical costs, but do not imply competitive constants at moderate budgets. In the numerical study the finer checkpoints never produced a determinate label, and the coarser configuration was worse than the retained-history identification adaptation in both displayed costs in two environments. Practical comparisons must therefore match the reporting task and required service level, as well as reward cost.

On the specified repeated-update family, rapid maintenance has matching expected-regret and joint-quantile orders. The deferred endpoint supplies an upper bound at a larger area budget. A matching deferred converse, the frontier at intermediate budgets, and achievable guarantees under general overlapping arrivals remain unresolved. Automatic precision selection also needs a valid confidence construction; choosing a favorable checkpoint after inspecting its observations is outside the fixed-precision guarantee. Mechanism experiments that separate evidence reuse from service intensity and boundary-band work would help explain finite-budget performance.

The supplement gives compatible-window truncations and a general information-allocation relaxation as necessary conditions, together with intermediate service configurations. It also proves eventual exact classification for a separate revisable, multiresolution procedure when arrivals stop, the final arm set is finite, and no arm lies on the membership boundary. That consistency statement has different assumptions and does not apply to fixed checkpoints. Contextual information, delayed observations, and changing reward laws would require further analysis.

Data and code availability

The accompanying project contains the simulation configuration, 5,760 run-level records, shared-stream construction, implementation, and plotting code used in this article. The configuration and recorded source hashes identify the experiment versions; computational details are given in the

supplement. These are simulated data. A public repository or permanent archive URL has not been specified in this manuscript.

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